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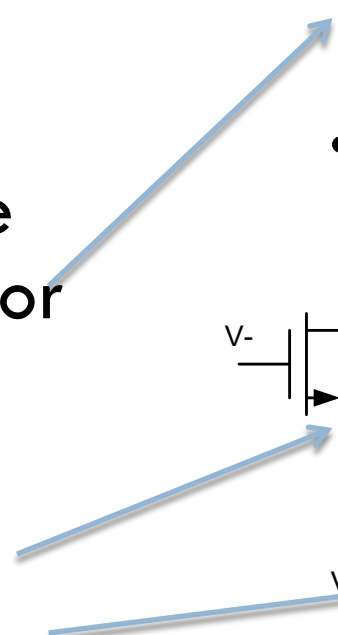
DESIGN OF OPAMPS

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Nome della conferenza o seminario

OPAMP general Specifications

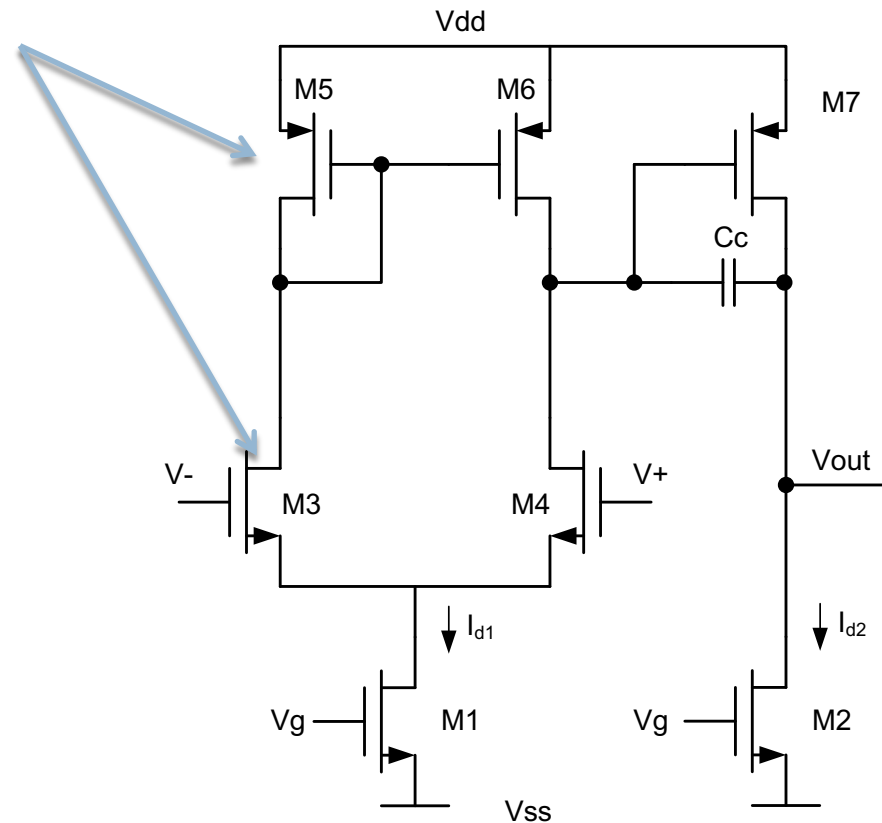
- DC Gain $A_v(0)$
 - ▣ *Also the (depend on the application) CMRR*
- Gain band product
- Input offset V_{offset}
- Input common mode range $ICMR$
- Output capacitance C_L
- Slew rate SR
- Output voltage swing $V_{out_{max}}$ $V_{out_{min}}$
- Consumption P_{diss}
- Input equivalent noise

- Depending on the ICMR spec choice of the n-ch differential stage or the p-ch
- Determination of the V_{ov} of M_5 and M_6 for the Max ICMR
- Determination of the V_{ov} of $M_{1,2,3,4}$ Min ICMR



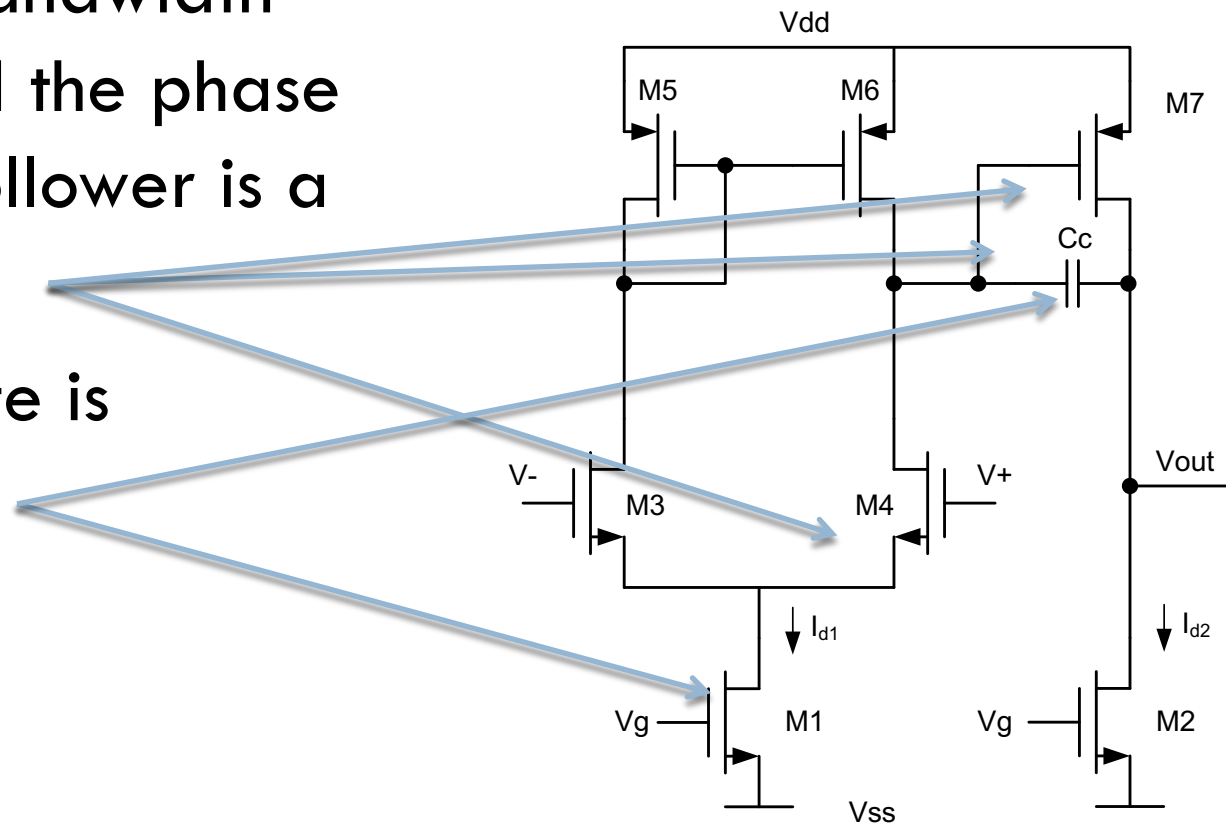
Two stage OPAMP design Steps

- The Input offset determine the dimension of the stage



Two stage OPAMP design Steps

- The Gain bandwidth product and the phase margin in follower is a function of
- The slew rate is function of



Project of two stage OpAmp

□ Determination of the capacitor C_c

$$GB \cong \frac{gm_{3,4}}{C_c}$$

$$p_1 \cong \frac{-1}{\left(\frac{r_{o7} r_{o2}}{r_{o7} + r_{o2}}\right) gm_7 \left(\frac{r_{o6} r_{o4}}{r_{o6} + r_{o4}}\right) C_c}$$

$$p_2 \cong \frac{-gm_7}{C_L}$$

$$z_1 = \frac{gm_7}{C_c}$$

$$60 < 180 - \tan^{-1}\left(\frac{GB}{p_1}\right) - \tan^{-1}\left(\frac{GB}{p_2}\right) - \tan^{-1}\left(\frac{GB}{z_1}\right)$$

Considering the ratio
0.1

$$gm_7 > 10 gm_{3,4}$$

$$|p_2| > 2.2 GB$$

$$\frac{gm_7}{C_L} > 2.2 \frac{gm_{3,4}}{C_c}$$

$$C_c > 0.22 C_L$$

Project of two stage OpAmp

- Determination of $gm_{3,4}$

$$gm_{3,4} \cong GB C_c$$

- From slew rate can be determined the I_{d1}

$$I_{d1} = SRCc$$

- From the two information can be calculated the $V_{ov_{3,4}}$

$$V_{ov_{3,4}} = \frac{2I_D}{gm_{3,4}} = \frac{I_{d1}}{gm_{3,4}}$$

Project of two stage OpAmp

- Determination of the ratio of 3,4

$$\left(\frac{W}{L}\right)_{3,4} = \frac{2\left(\frac{I_{d1}}{2}\right)}{K'_n [V_{ov_{3,4}}]^2}$$

- As a function of the ICMR max the $(W/L)_{5,6}$ is determined

$$\left(\frac{W}{L}\right)_{5,6} = \frac{2\left(\frac{I_{d1}}{2}\right)}{K'_p [V_{DD} - V_{in}(\max) - |V_{T5}|(\max) + V_{T3}(\min)]^2}$$

Project of two stage OpAmp

- As a function of the offset we will determine the values of Area

Minimum area required

$$\sigma_{Vos} = \sqrt{\sigma_{V_{tp3,4}}^2 + \left(\frac{V_{OV3,4}}{2}\right)^2 \left[\left(\frac{\sigma_{Kp}}{k_p}\right)_{3,4}^2 + \left(\frac{\sigma_{I_D}}{I_D}\right)_{5,6}^2 \right]}$$

$$\left(\frac{\sigma_{I_D}}{I_D}\right)_{5,6} = \sqrt{\left(\frac{\sigma_{Kp}}{k_p}\right)_{5,6}^2 + \left(\frac{2\sigma_{V_{Tp5,6}}}{V_{OV5,6}}\right)^2}$$

$W_{3,4,5,6}$

$L_{3,4,5,6}$

Project of two stage OpAmp

- As a function of the ICMR min the $(W/L)_1$ is determined

$$V_{in\min} = V_{ov1} + V_{ov_{3,4}} + V_{T\max}$$



$$\left(\frac{W}{L}\right)_1 = \frac{2(I_{d1})}{K'_n [V_{ov1}]^2}$$

Project of two stage OpAmp

- Considering the transconductance relation

$$gm_7 > 10 gm_{3,4}$$

- Considering

$$V_{ov_{5,6}} = V_{ov_7}$$

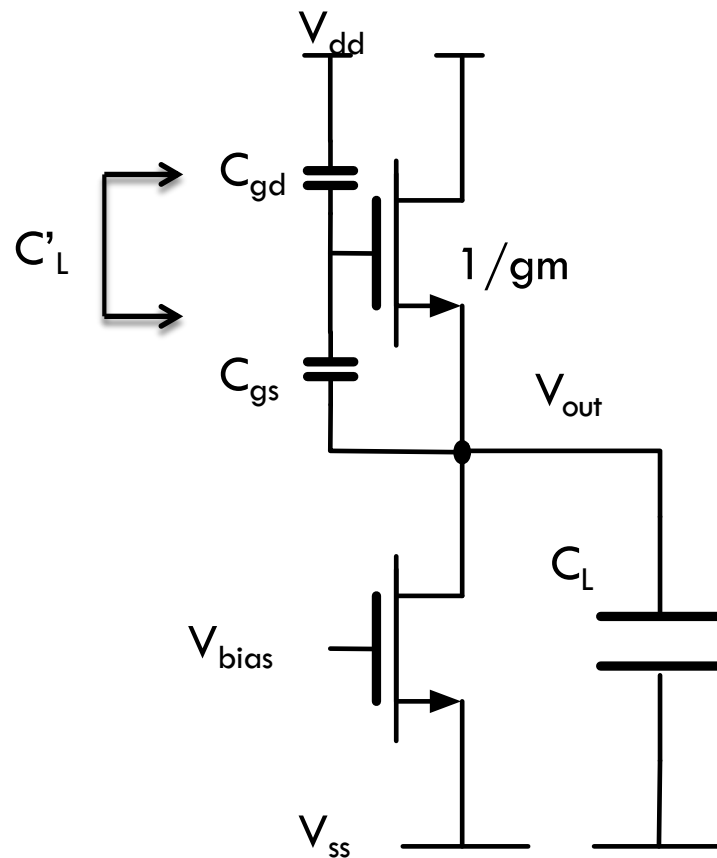
$$\left(\frac{W}{L}\right)_7 > \frac{10 gm_{3,4}}{K_P V_{ov_{5,6}}}$$

$$Id_7 = Id_2 = \left(\frac{W}{L}\right)_7 / \left(\frac{W}{L}\right)_{5,6} \frac{Id_1}{2}$$

$$\left(\frac{W}{L}\right)_2 = \left(\frac{W}{L}\right)_1 \frac{Id_2}{Id_1}$$

Output stage

- Obviously a buffer output stage reduce C_c



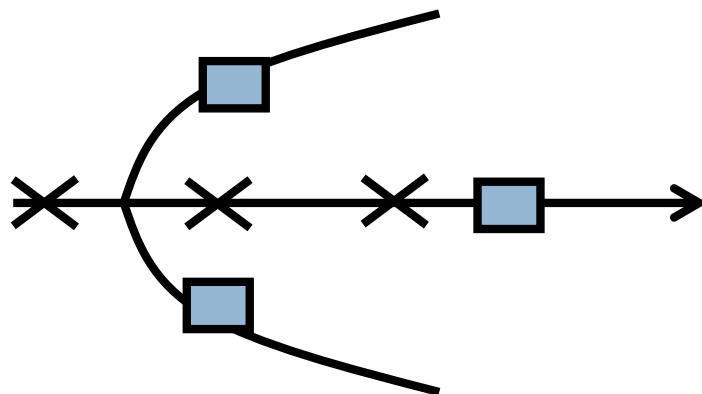
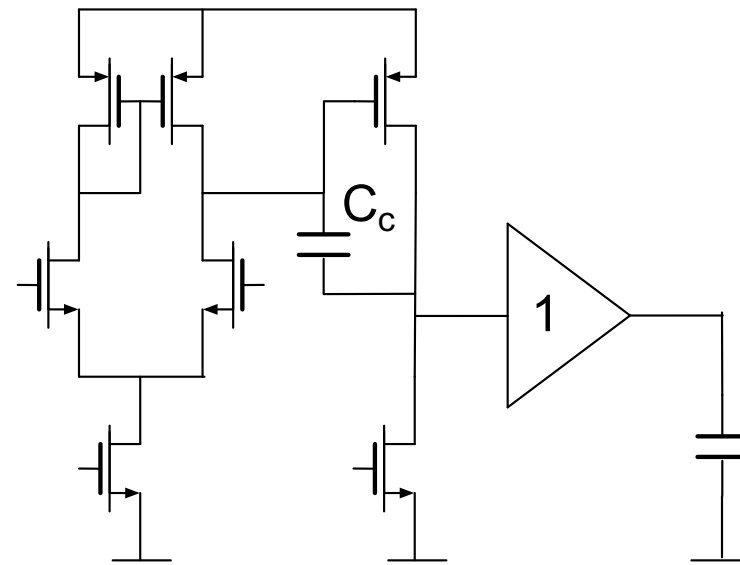
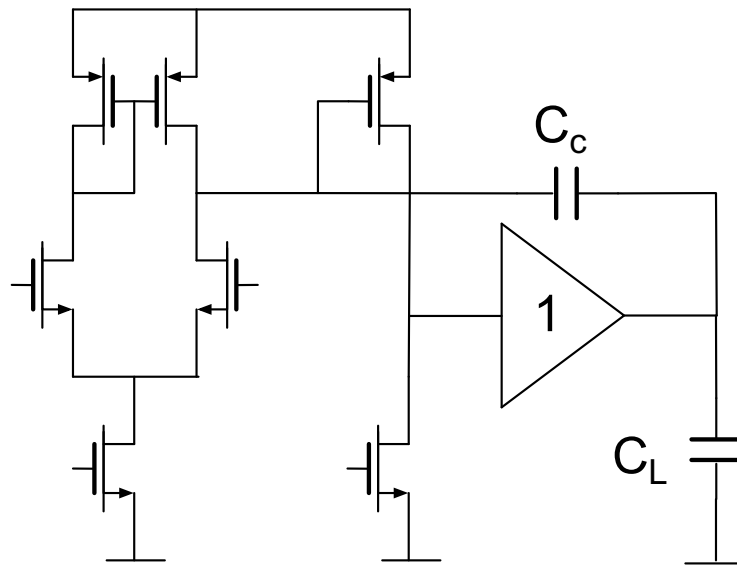
$$C'_L < C_L$$

Condition to be imposed

$$|p_3| \cong \frac{1}{R_{out} C_L} = \frac{gm_{out}}{C_L} > 10GB$$

Output stage

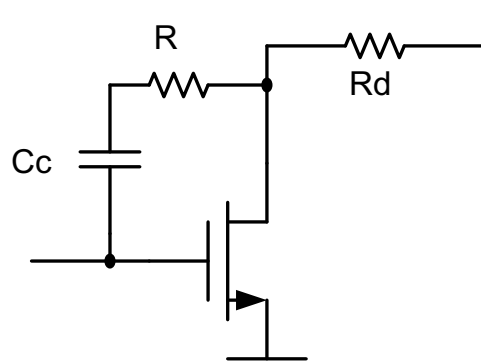
□ Output stage



Inserting a zero on the loop transfer function

- Inserting the zero in the transfer function

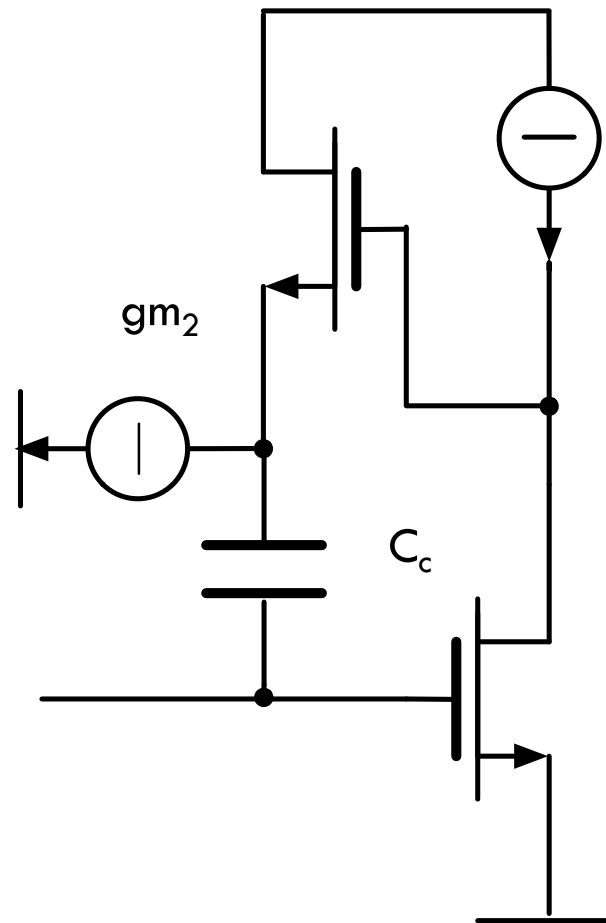
Second Stage of the OpAmp



$$z_1 = \frac{1}{C_c (1/gm - R)}$$

Inserting a zero on the loop transfer function

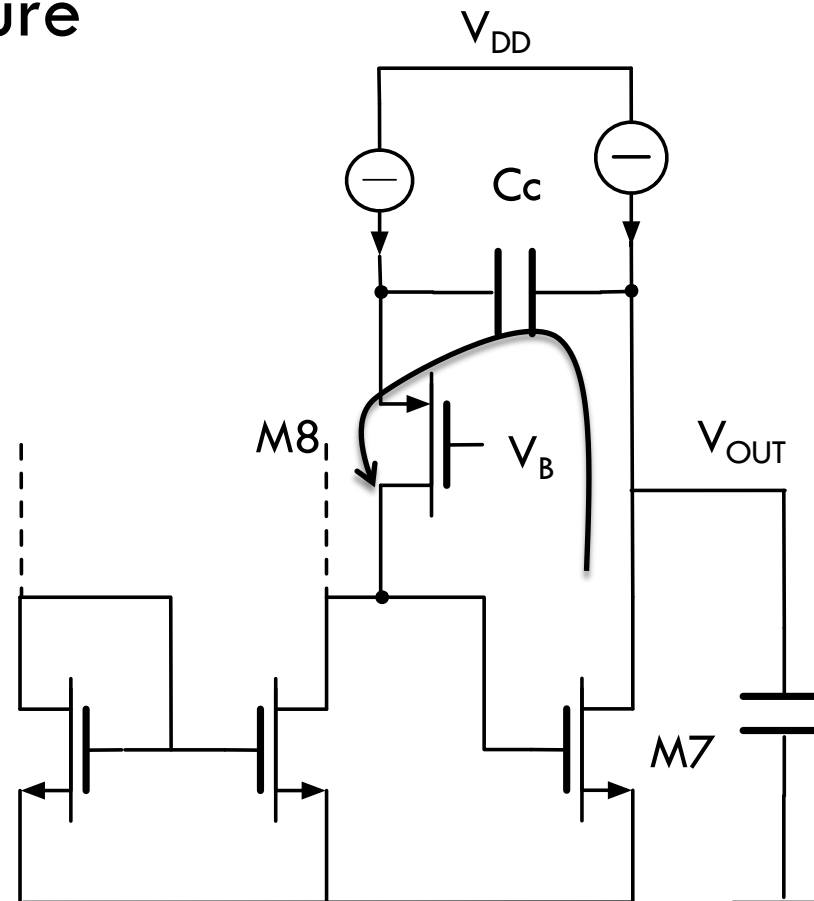
- Controlling the right half-plane zero



$$z_1 = -\frac{1}{C_c (1/gm_2)}$$

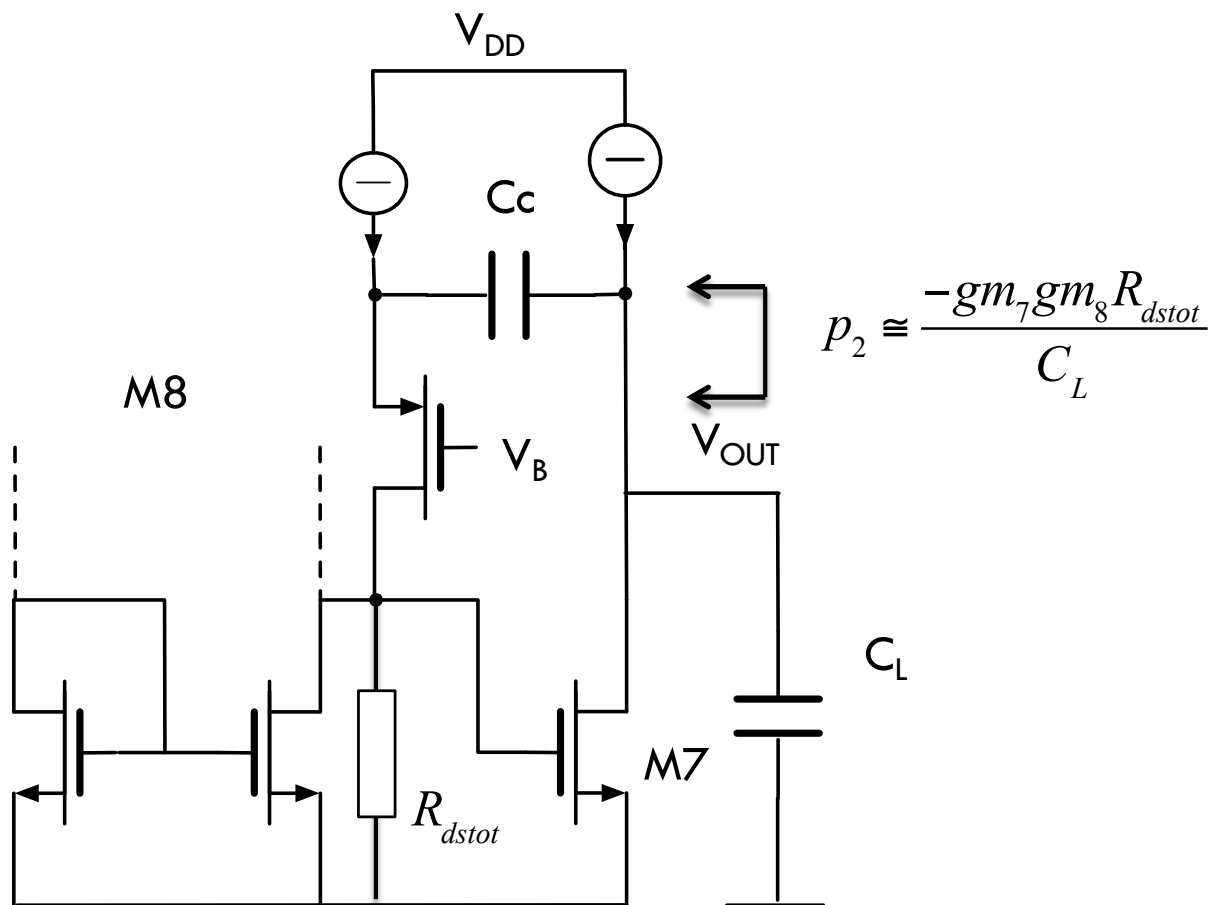
Compensation that increase p_2

- The feedback gain reduce the output impedance of the structure



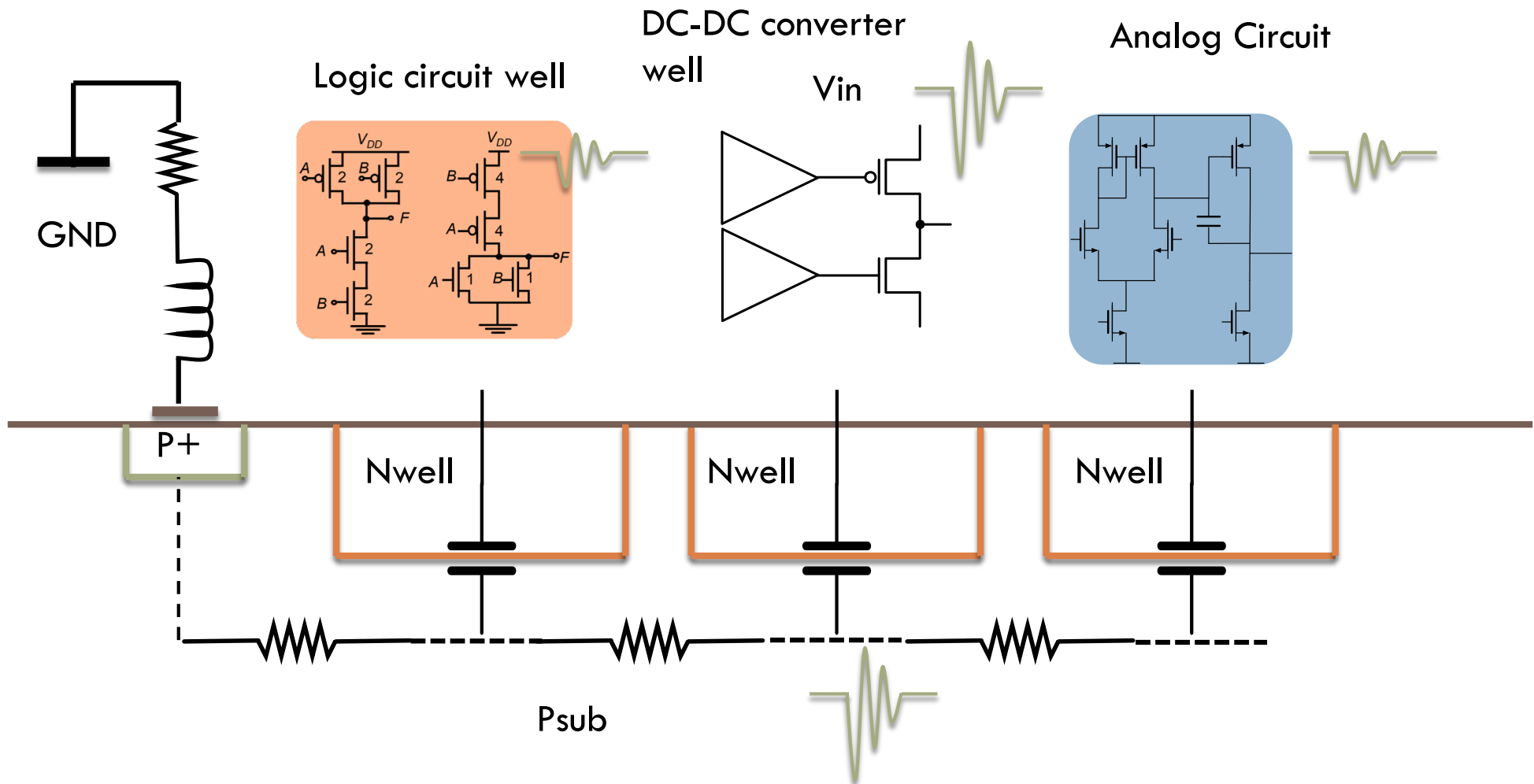
Compensation that increase p_2

- At high frequency



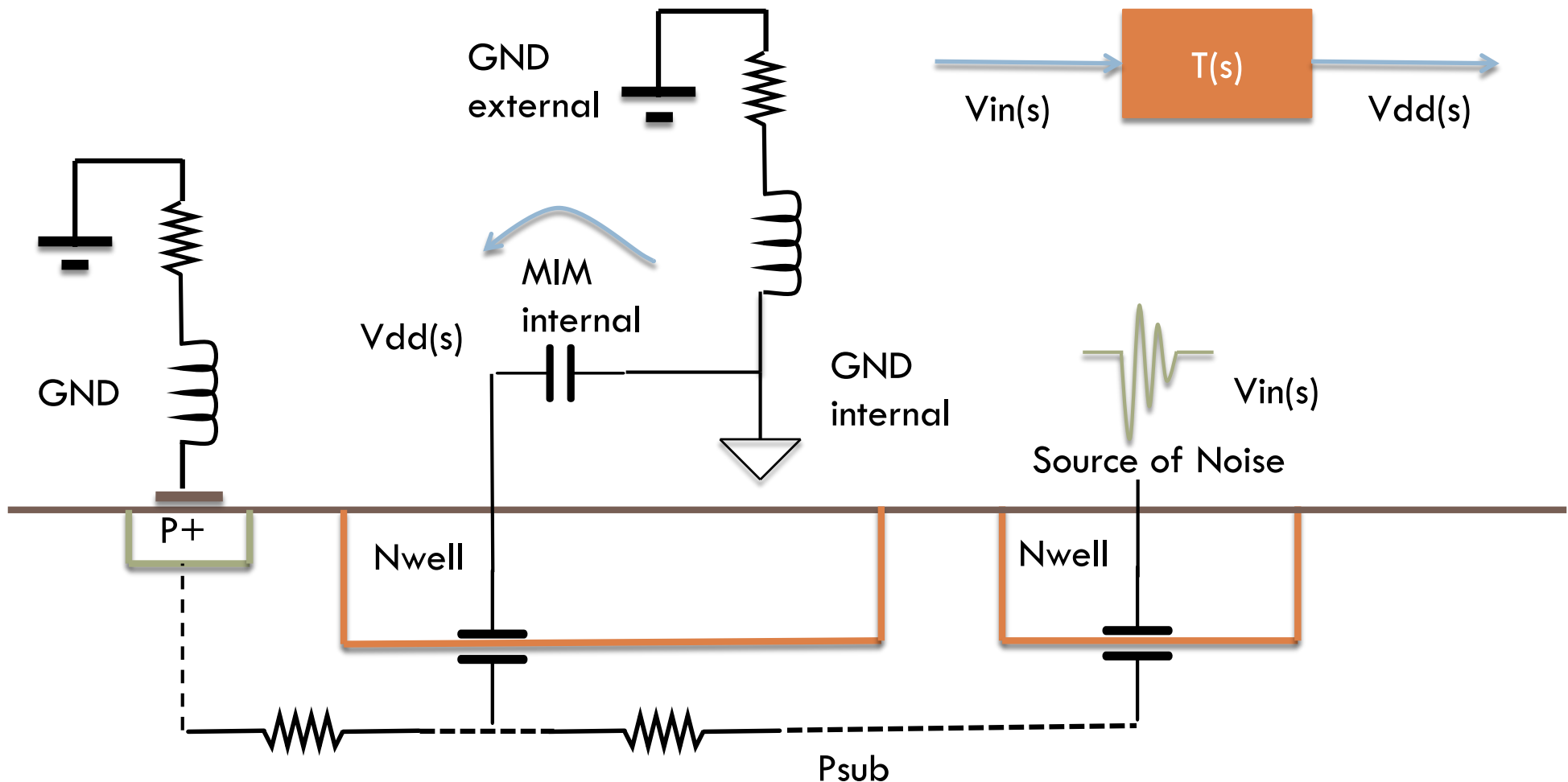
PSRR: Substrate noise coupling

□ Vdd and Gnd bounce



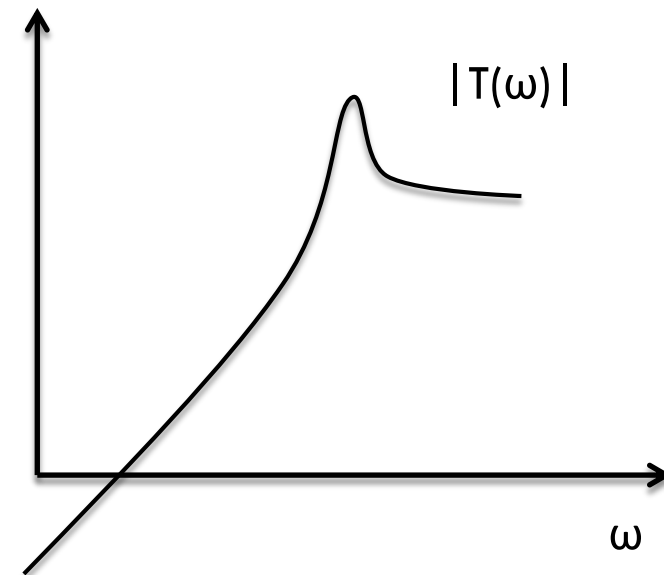
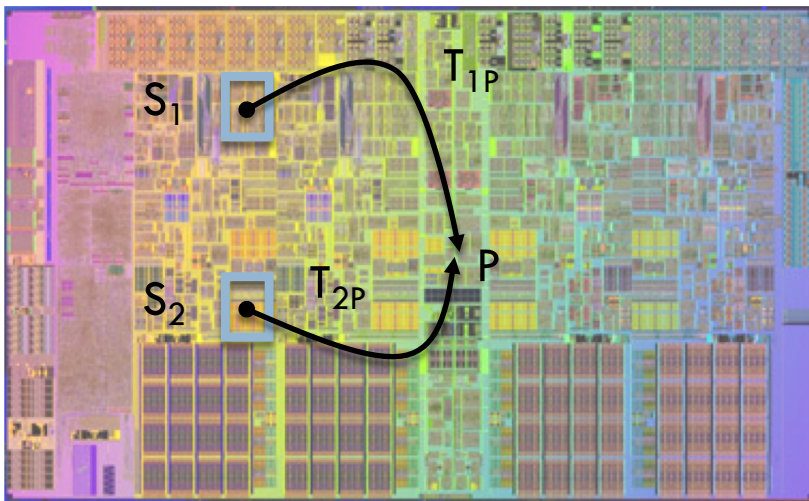
PSRR: Substrate noise coupling Analysis

- For each point can be defined a $T(s)$



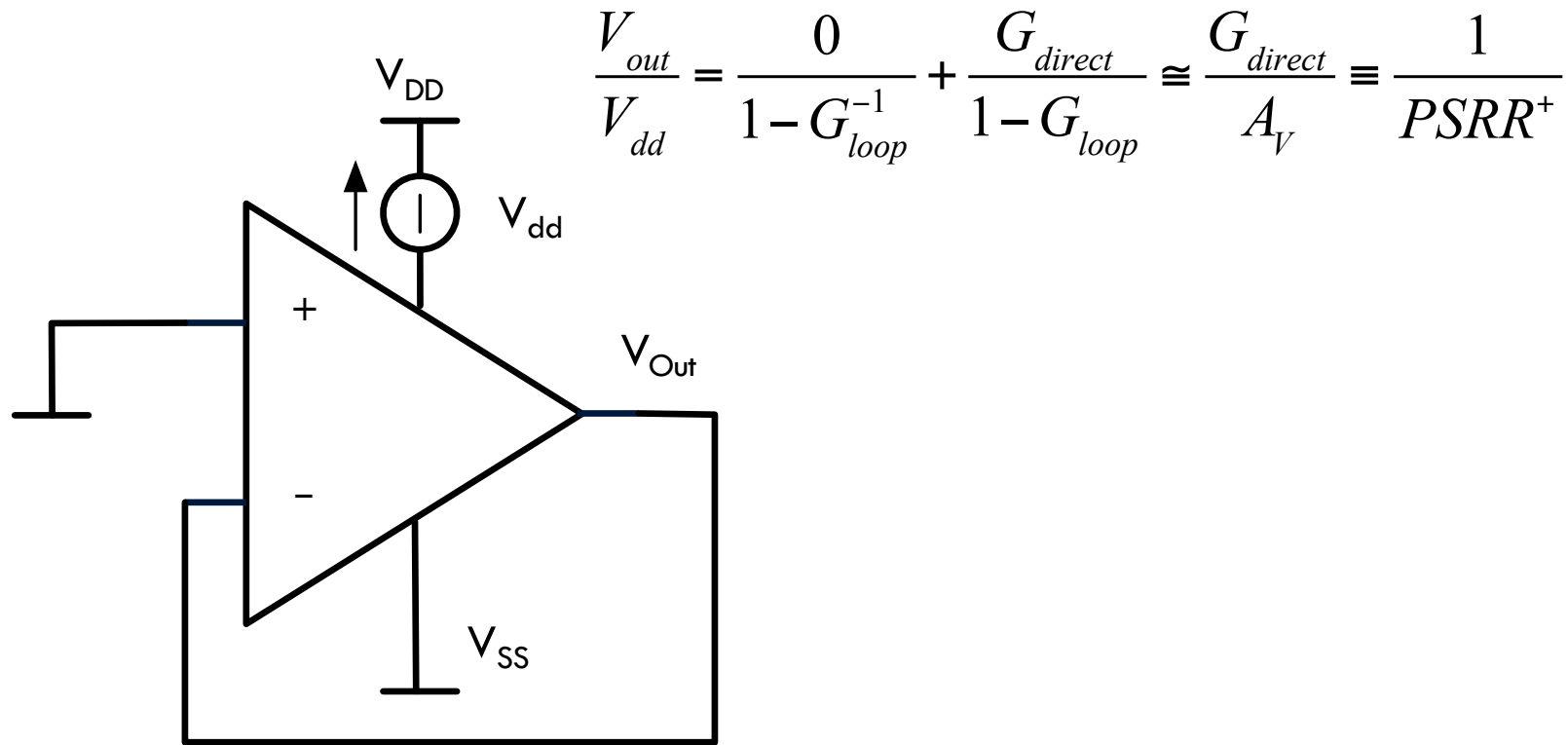
PSRR: Substrate noise coupling Analysis

- For each sensitive point and each noise source can be defined a $T(s)$



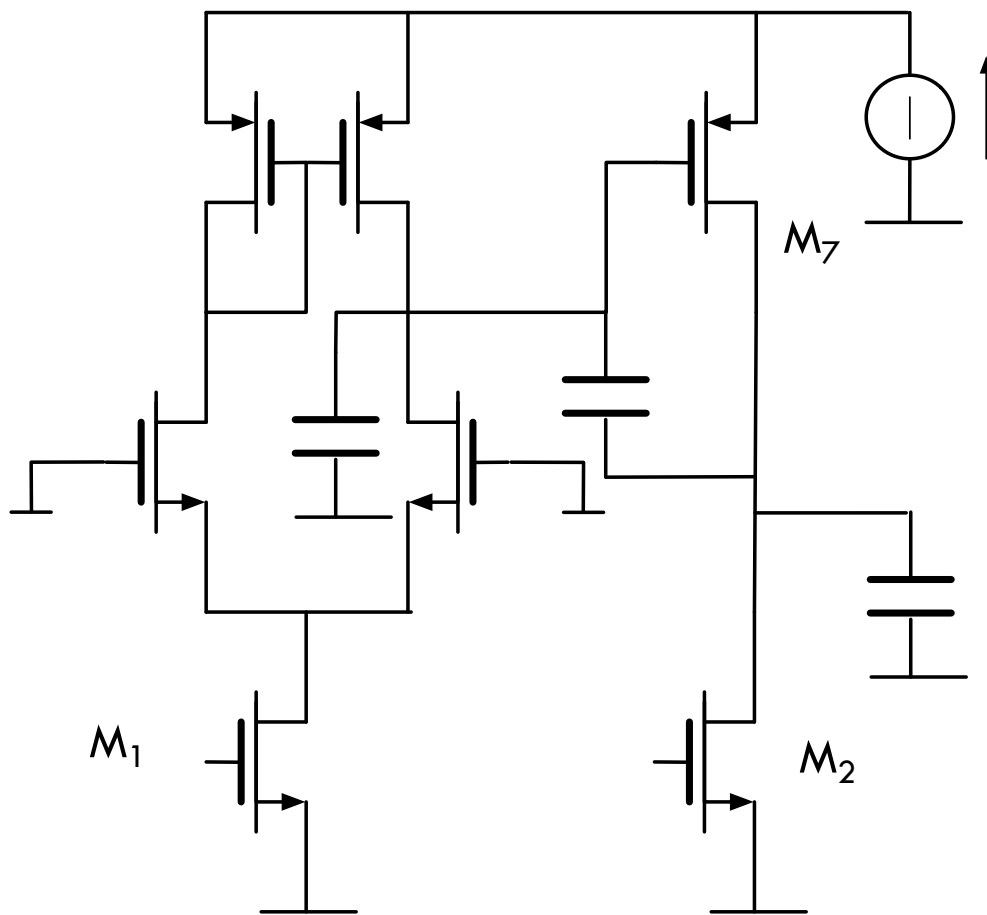
PSRR

□ Definition

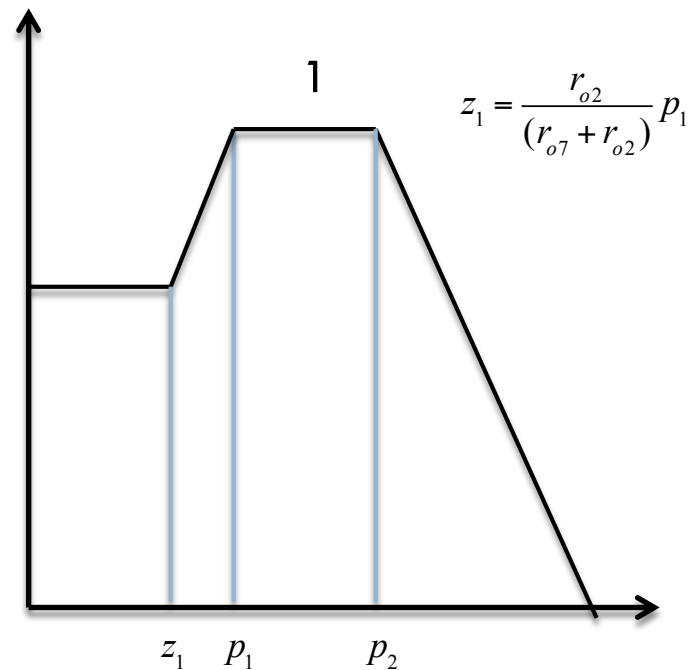


PSRR

□ Calculation

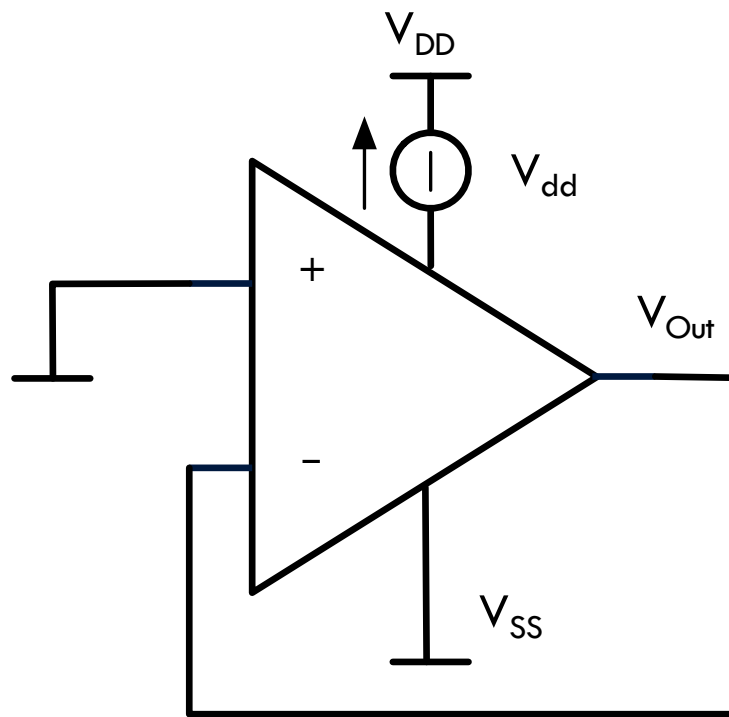


$$G_{diretto} = \frac{r_{o2}}{(r_{o7} + r_{o2})} \frac{(1 + s\tau_z)}{(1 + s\tau_1)(1 + s\tau_2)}$$

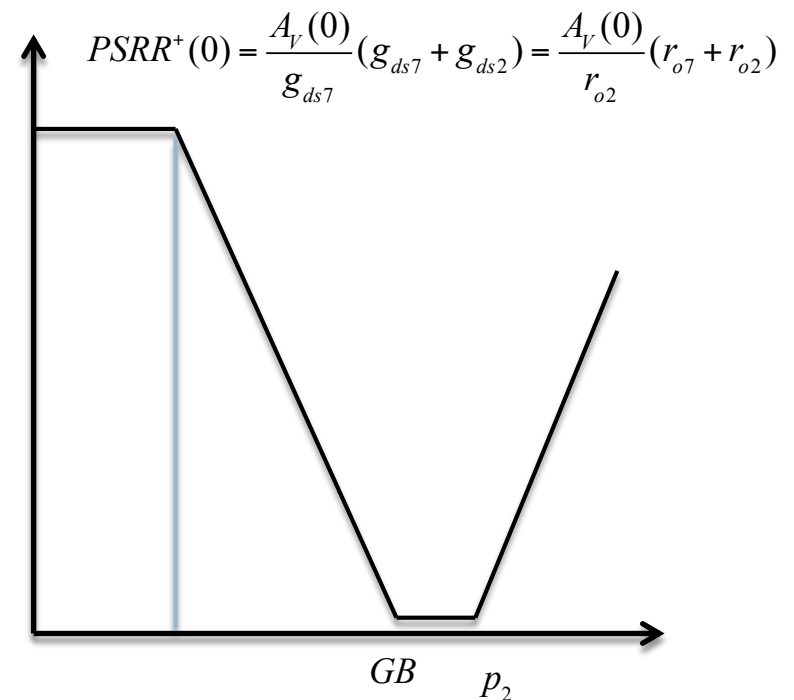


PSRR close loop

□ Calculation close loop

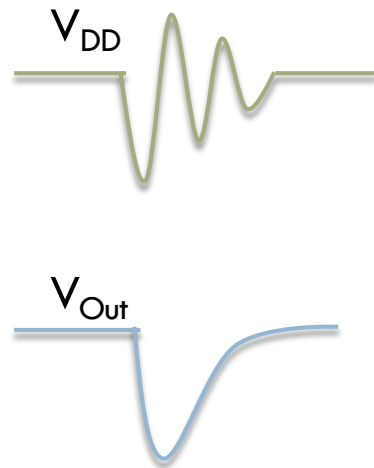


$$PSRR^+ = \frac{v_{dd}}{v_{out}} = \frac{1 - G_{loop}}{G_{direct}}$$



PSRR

- Non Linear effect of the OpAmp as Asymmetrical slew rate can rectify the noise
- This effect are studied by AC simulations



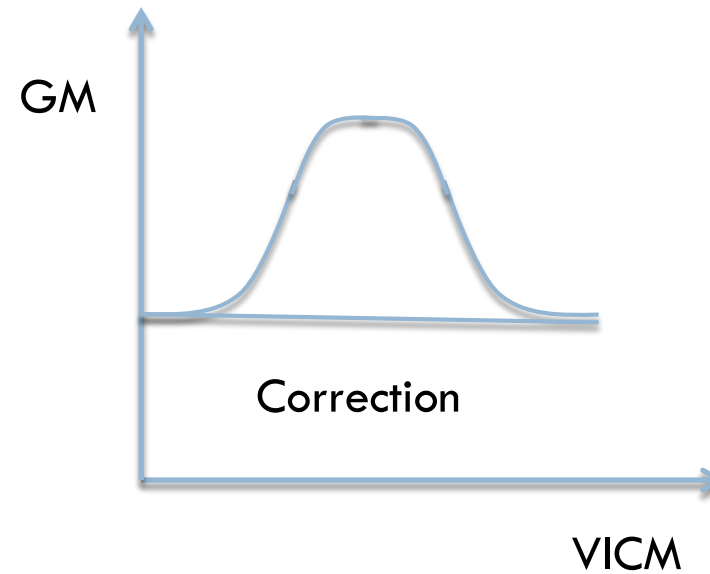
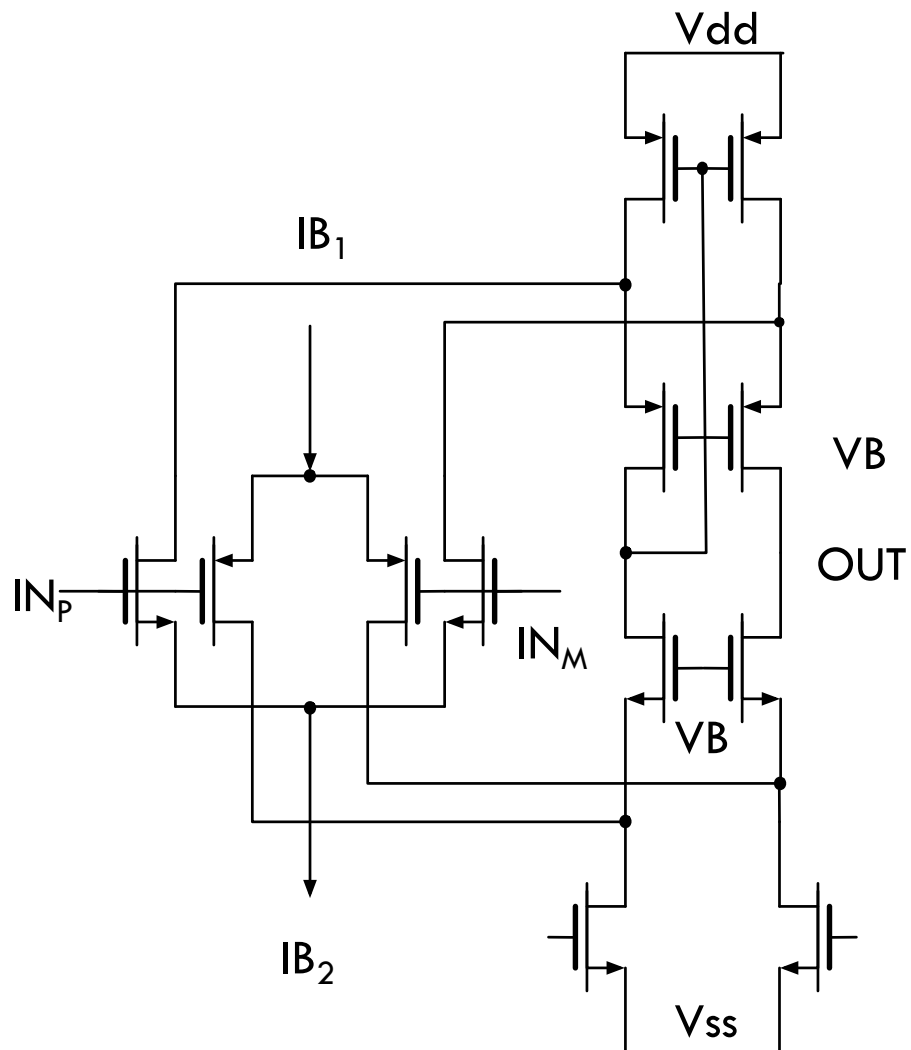
Low Noise design

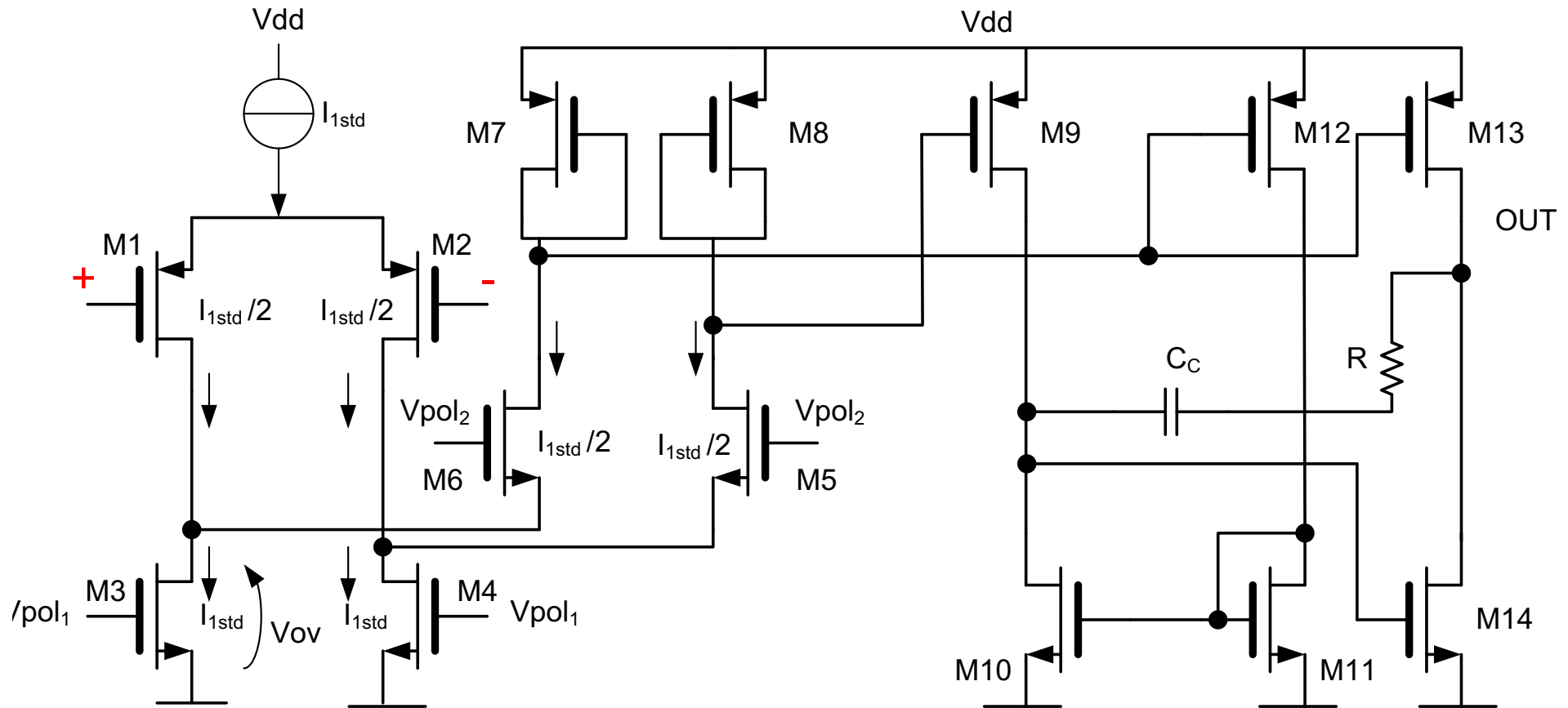
- The PMOS have about 2 to 5 less $1/f$ noise than n
 - Design a PMOS input stage
- Make the first gain stage as high as possible
 - Cascode structure to increase the low frequency gain of the opamp

- Increase the Input common mode range

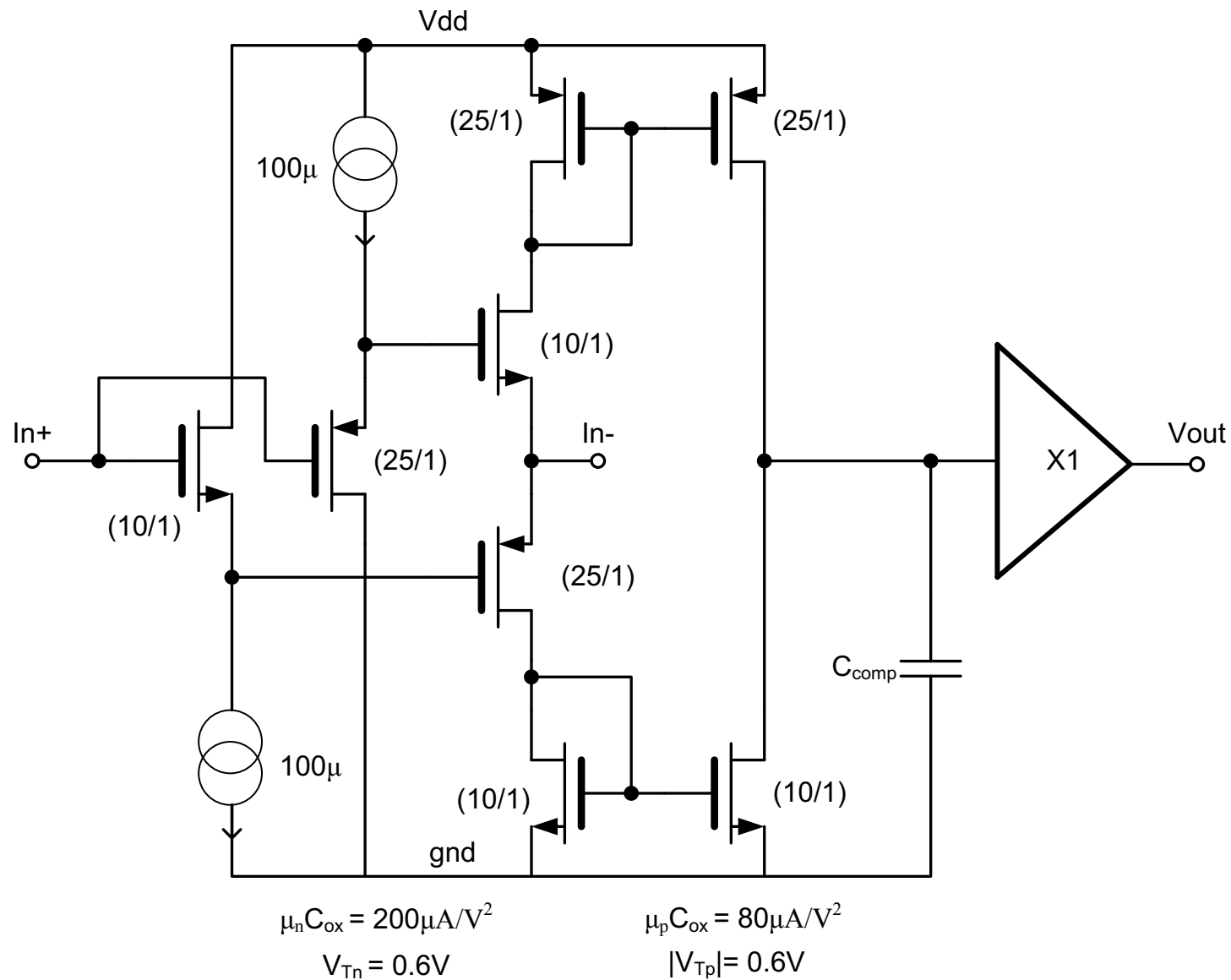


Input rail to rail

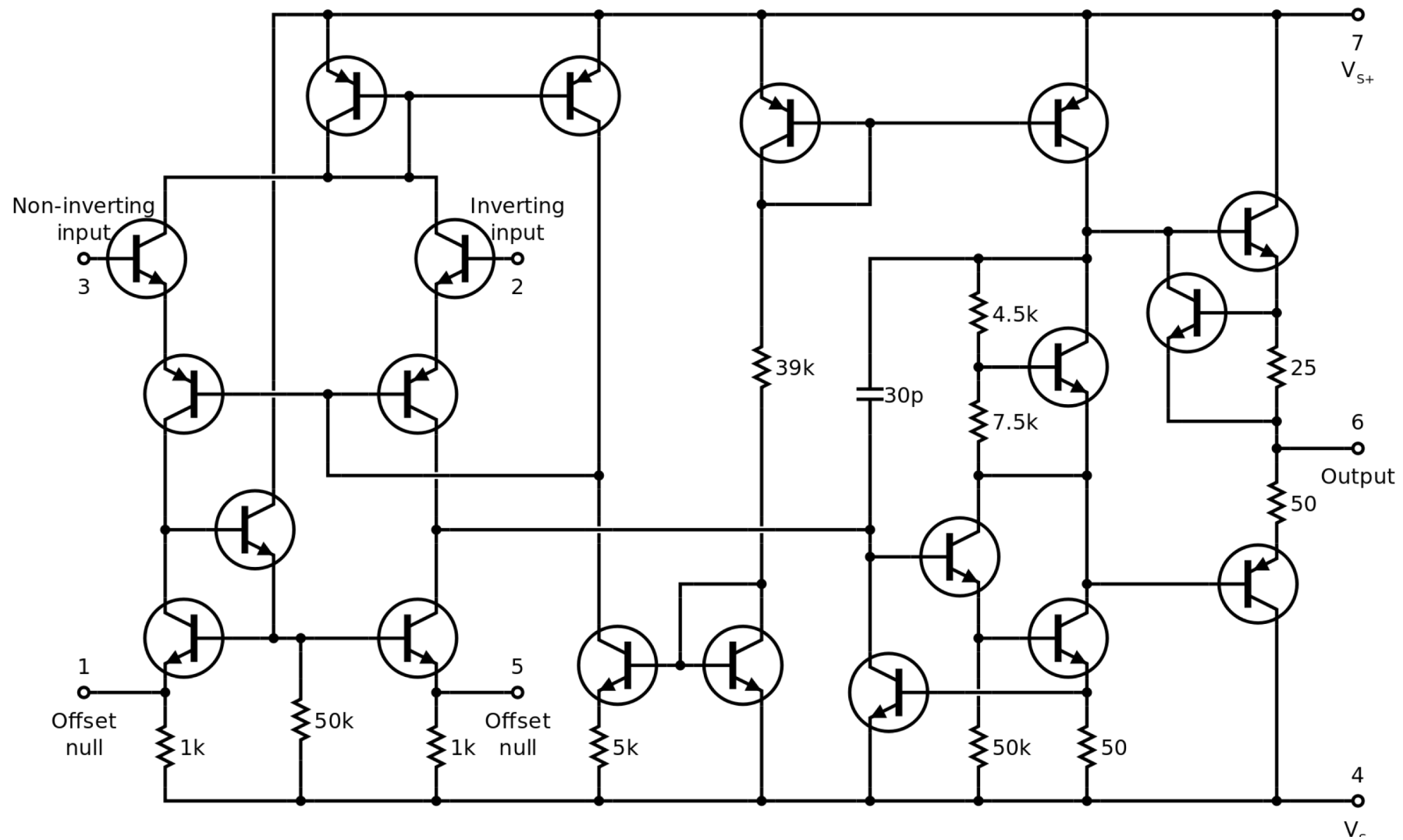




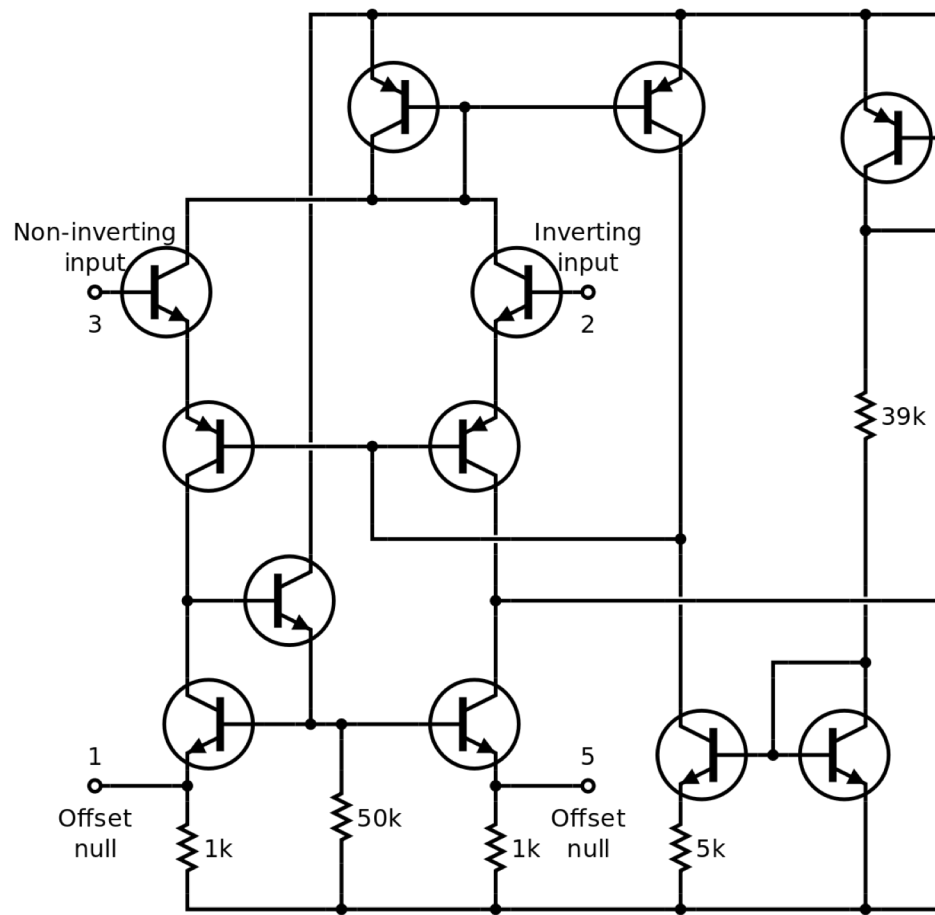
Current feedback



$\mu 741$



Input stage



- Input stage *n*pn higher β
- Gain $g_m/4$ of differential stage gain final $g_m/2$
- Mirror with offset compensation
- Second stage *n*pn higher performance
- Bias loop for first stage

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